

The empirical recurrence for OEIS A257266: recovering a conjecture that is not written down

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Abstract

OEIS A257266 records “Empirical recurrence of order 48”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 113$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 48, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 48 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A257266 is “Number of $(n+2) \times (1+2) \dots 2$ arrays with no 3×3 subblock diagonal sum less than the antidiagonal sum or central row sum equal to the central column sum.”. It has offset 1 and begins

9300, 119716, 1257600, 13089924, 121905264, 1125468304, 9711591328, 83266873600, ...

The entry states:

Empirical recurrence of order 48 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 15:20:46, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 757$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 113$ of the 757, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 113$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1514$ exact terms of the model returns order 48 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{48} c_i a(n-i),$$

c_1	18	c_{13}	−13849777152	c_{25}	−5428374780542976	c_{37}	−7951752463830220800
c_2	34	c_{14}	43038111744	c_{26}	8227910627131392	c_{38}	11690378641303142400
c_3	−2352	c_{15}	36299003904	c_{27}	−2182500041490432	c_{39}	−14120298253438156800
c_4	7400	c_{16}	−810119996928	c_{28}	−60804406832136192	c_{40}	−5962766009499648000
c_5	117072	c_{17}	1456628825088	c_{29}	146806760456847360	c_{41}	43881145612369920000
c_6	−675536	c_{18}	3774832118784	c_{30}	61801088955973632	c_{42}	−29268028481863680000
c_7	−2498880	c_{19}	−16606473043968	c_{31}	−501643577167183872	c_{43}	−9161293210583040000
c_8	24229952	c_{20}	29286163132416	c_{32}	614182603092590592	c_{44}	29478857107046400000
c_9	5154048	c_{21}	15848098799616	c_{33}	−376194649561104384	c_{45}	−60518031792537600000
c_{10}	−416571392	c_{22}	−410789346435072	c_{34}	−1863213747449757696	c_{46}	26191045066752000000
c_{11}	787840512	c_{23}	849862304464896	c_{35}	5493893835864932352	c_{47}	46809527353344000000
c_{12}	2141033472	c_{24}	1083613196795904	c_{36}	−1375736761645793280	c_{48}	−33435376680960000000

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 48, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $48 + 4$ terms removed returns order 48 as well, so $\deg q_{\min} = 48 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $48 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 48 that this sequence satisfies — and that family turns out to have exactly one member.

References

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